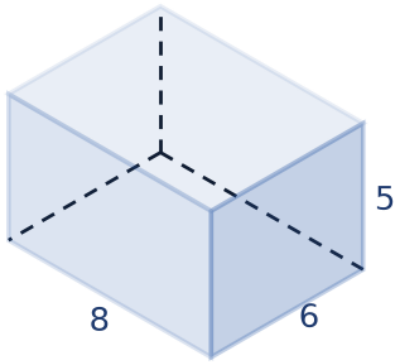
# Area, Volume, 3D Solids & Modeling Strategy

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

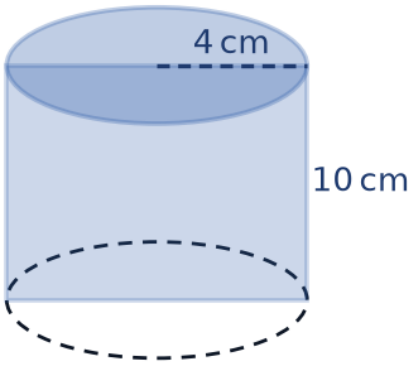
**1** [Foundational] • [FRQ] • Target: Rectangular Prism Volume, Surface Area & Space Diagonal

A rectangular prism has length $l = 8\,\text{cm}$, width $w = 6\,\text{cm}$, and height $h = 5\,\text{cm}$, as shown. Find (a) its volume, (b) its total surface area, and (c) the exact length of its space diagonal $d$.




**2** [Foundational] • [FRQ] • Target: Cylinder Volume & Surface Area

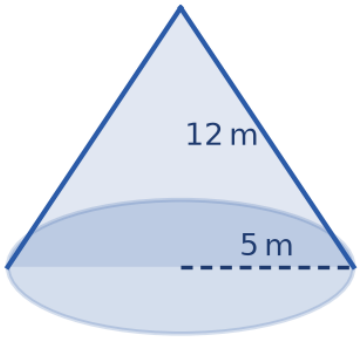
A right circular cylinder has base radius $r = 4\,\text{cm}$ and height $h = 10\,\text{cm}$. Compute its exact volume, lateral surface area, and total surface area in terms of $\pi$.




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**3** [Foundational] • [FRQ] • Target: Right Circular Cone Geometry

A right circular cone has base radius $r = 5\,\text{m}$ and vertical height $h = 12\,\text{m}$. Find (a) its slant height $\ell$, (b) its exact volume, and (c) its total surface area in terms of $\pi$.

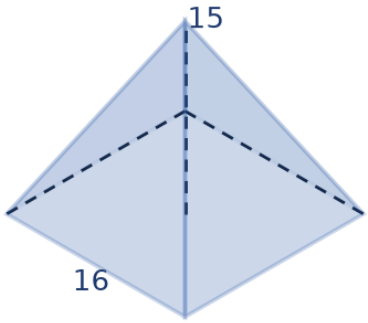



**4** [Foundational] • [FRQ] • Target: Sphere Surface Area & Volume

A sphere has surface area $144\pi\,\text{cm}^2$. Find its radius $r$ and its exact volume $V$ in terms of $\pi$.

**5** [Foundational] • [FRQ] • Target: Square Pyramid Volume & Slant Height

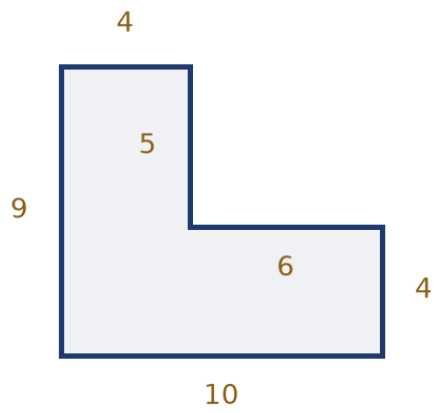
A right square pyramid has base edge length $b = 16\,\text{cm}$ and vertical height $h = 15\,\text{cm}$. Find (a) its volume, (b) the slant height $\ell$ of each triangular face, and (c) its total surface area.




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**6** [Foundational] • [FRQ] • Target: Composite L-Shaped Polygons

Compute the area and perimeter of the L-shaped polygon with vertices $(0, 0), (10, 0), (10, 4), (4, 4), (4, 9), (0, 9)$.




**7** [Foundational] • [FRQ] • Target: Rhombus Diagonals, Area & Perimeter

A rhombus has diagonals of lengths $d_1 = 16\,\text{cm}$ and $d_2 = 30\,\text{cm}$. Find the area and perimeter of the rhombus.

## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Volume Conservation Across Recast Solids

A solid metal sphere of radius $6\,\text{cm}$ is melted down and recast into a solid right circular cone of base radius $12\,\text{cm}$. Find the height $h$ of the cone.

**9** [Intermediate] • [FRQ] • Target: Annular / Hollow Cylinder Volume

A cylindrical pipe of length $20\,\text{cm}$ has outer radius $R = 5\,\text{cm}$ and inner hollow radius $r = 3\,\text{cm}$. Find the exact volume of metal in the pipe wall.

**10** [Intermediate] • [FRQ] • Target: Composite 3D Solids (Cylinder + Hemisphere)

A grain silo consists of a right circular cylinder of radius $3\,\text{m}$ and height $10\,\text{m}$ topped by a hemispherical dome of radius $3\,\text{m}$. Find the exact total volume of the silo in terms of $\pi$.

**11** [Intermediate] • [FRQ] • Target: Inscribed Sphere in a Cube

A sphere is inscribed inside a cube of edge length $8\,\text{cm}$ so that the sphere is tangent to all $6$ faces. Find the exact volume of the empty space inside the cube but outside the sphere.

**12** [Intermediate] • [FRQ] • Target: Net of a Cone from a Circular Sector

A circular sector of radius $10\,\text{cm}$ and arc length $12\pi\,\text{cm}$ is rolled up to form the lateral surface of a right circular cone (without overlap). Find the base radius $r$ and vertical height $h$ of the resulting cone.

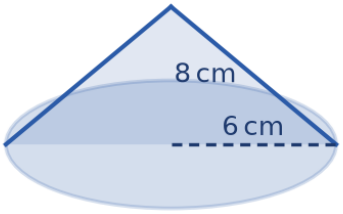
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**13** [Intermediate] • [FRQ] • Target: Hexagonal Prism Volume & Surface Area

A regular hexagonal prism has base side length $4\,\text{cm}$ and height $10\,\text{cm}$. Find its exact volume and total surface area.

**14** [Intermediate] • [FRQ] • Target: Solids of Revolution of Polygons

A right triangle with legs $6\,\text{cm}$ and $8\,\text{cm}$ is revolved $360^\circ$ about its $8\,\text{cm}$ leg. Describe the resulting 3D solid and compute its exact volume and lateral surface area.



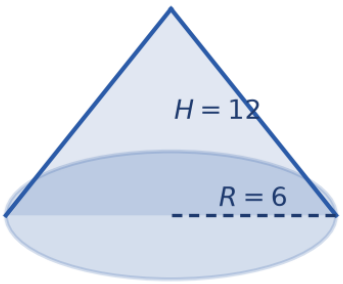

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • [FRQ] • Target: 3D Box Construction via Quadratic Equations • Cross-Lesson: Topic 3.01 (Quadratics)

An open-top rectangular box is constructed by cutting congruent squares of side $2\,\text{cm}$ from the four corners of a rectangular sheet of tin whose length is $5\,\text{cm}$ greater than its width, and folding up the flaps. If the volume of the box is $100\,\text{cm}^3$, find the original dimensions of the tin sheet.

**16** [Challenging] • [FRQ] • Target: Similar Cones & Cubic Volume Ratio • Cross-Lesson: Topic 7.01 (Similar Triangles) & Topic 5.01 (Ratios)

A conical water tank (vertex pointing down) has top radius $R = 6\,\text{m}$ and total height $H = 12\,\text{m}$. When the water in the tank has depth $h = 8\,\text{m}$, find the radius $r$ of the water surface and the fraction of the tank's total volume that is filled.




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**17** [Challenging] • [FRQ] • Target: Cube Inscribed in a Sphere • Cross-Lesson: Topic 7.01 (3D Pythagorean Theorem)

A cube is inscribed inside a sphere of radius $R = 3\sqrt{3}$ cm so that all $8$ vertices of the cube lie on the surface of the sphere. Find the edge length $s$ and volume $V$ of the cube.

**18** [Challenging] • [FRQ] • Target: Cylinder Inscribed in a Sphere • Cross-Lesson: Topic 9.01 (Circles & Chords)

A right circular cylinder is inscribed in a sphere of radius $5\,\text{cm}$. If the height of the cylinder is $6\,\text{cm}$, find the base radius $r$ and exact volume of the cylinder.

**19** [Challenging] • [FRQ] • Target: Volumetric Flow Rate & Dimensional Modeling • Cross-Lesson: Topic 5.01 (Rates & Proportions)

Water flows through a cylindrical pipe of internal diameter $4\,\text{cm}$ at a linear speed of $50\,\text{cm/s}$ into an empty rectangular tank of base $100\,\text{cm}$ by $60\,\text{cm}$. How many seconds (in terms of $\pi$) does it take to fill the tank to a depth of $20\,\text{cm}$?

**20** [Challenging] • [FRQ] • Target: 3D Regular Tetrahedron Altitude & Volume • Cross-Lesson: Topic 7.01 (Equilateral Triangles & Centroids)

Find the exact surface area and volume of a regular tetrahedron (all $4$ faces are equilateral triangles) of edge length $s = 6\,\text{cm}$.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Cube Surface Area to Volume

What is the volume of a cube whose total surface area is $150\,\text{cm}^2$?

**a.** $25\,\text{cm}^3$

**b.** $100\,\text{cm}^3$

**c.** $125\,\text{cm}^3$

**d.** $216\,\text{cm}^3$

**22** [Foundational] • [MCQ] • Target: Cylinder Volume & Base Circumference

A right circular cylinder has volume $72\pi$ and height $8$. What is the circumference of its circular base?

**a.** $3\pi$

**b.** $6\pi$

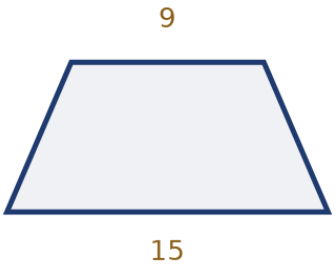
**c.** $9\pi$

**d.** $12\pi$

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**23** [Foundational] • [MCQ] • Target: Area of a Trapezoid

What is the area of a trapezoid with parallel bases of lengths $9$ and $15$ and height $7$?




**a.** 42

**b.** 84

**c.** 105

**d.** 168

**24** [Foundational] • [MCQ] • Target: Comparing Cone & Cylinder Volumes

A right circular cone and a right circular cylinder have the same base radius and the same volume. The height of the cone is how many times the height of the cylinder?

**a.** $\frac{1}{3}$

**b.** 1

**c.** 3

**d.** 9

**25** [Foundational] • [MCQ] • Target: Cubic Scaling of Sphere Volume

If the radius of a sphere is doubled, its volume is multiplied by what factor?

**a.** 2

**b.** 4

**c.** 6

**d.** 8

**26** [Foundational] • [MCQ] • Target: Rectangle Perimeter & Area

A rectangular field has perimeter $100\,\text{m}$ and length $32\,\text{m}$. What is the area of the field?

**a.** $544\,\text{m}^2$

**b.** $576\,\text{m}^2$

**c.** $640\,\text{m}^2$

**d.** $1024\,\text{m}^2$

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**27** [Foundational] • [MCQ] • Target: Volume of a Triangular Prism

What is the volume of a right triangular prism whose base is a right triangle with legs $5$ and $12$ and whose prism height is $10$?

**a.** 150

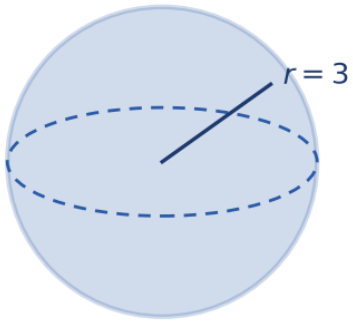
**b.** 300

**c.** 600

**d.** 650

**28** [Foundational] • [MCQ] • Target: Volume of a Sphere

A solid sphere of radius $r = 3$ is shown below. What is its exact volume?




**a.** $12\pi$

**b.** $27\pi$

**c.** $36\pi$

**d.** $108\pi$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

**29** [Intermediate] • [MCQ] • Target: Additive Volumes of Spheres ($3^3 + 4^3 + 5^3 = 6^3$)

Three solid lead spheres of radii $3\,\text{cm}$, $4\,\text{cm}$, and $5\,\text{cm}$ are melted together and cast into a single large sphere. What is the radius of the new sphere?

**a.** $6\,\text{cm}$

**b.** $7\,\text{cm}$

**c.** $8\,\text{cm}$

**d.** $12\,\text{cm}$

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**30** [Intermediate] • [MCQ] • Target: 3D Space Diagonal

What is the length of the longest straight rod that can fit completely inside a rectangular box measuring $3\,\text{ft}$ by $4\,\text{ft}$ by $12\,\text{ft}$?

**a.** $12\,\text{ft}$

**b.** $13\,\text{ft}$

**c.** $15\,\text{ft}$

**d.** $19\,\text{ft}$

**31** [Intermediate] • [MCQ] • Target: Archimedes Displacement in a Cylinder

A cylindrical glass of radius $3\,\text{cm}$ contains water to a depth of $8\,\text{cm}$. When a solid glass marble of radius $3\,\text{cm}$ is completely submerged in the water, by how many centimeters does the water level rise?

**a.** $2\,\text{cm}$

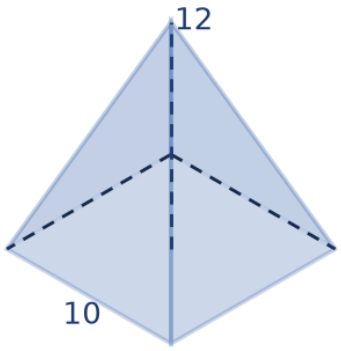
**b.** $3\,\text{cm}$

**c.** $4\,\text{cm}$

**d.** $6\,\text{cm}$

**32** [Intermediate] • [MCQ] • Target: Working Backward from Pyramid Volume

A square-based pyramid has volume $400\,\text{cm}^3$ and height $12\,\text{cm}$. What is the perimeter of its square base?




**a.** $20\,\text{cm}$

**b.** $40\,\text{cm}$

**c.** $80\,\text{cm}$

**d.** $100\,\text{cm}$

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**33** [Intermediate] • [MCQ] • Target: Cylinder Formed from a Rectangular Sheet

A rectangular sheet of paper of dimensions $6\pi$ cm by $10$ cm is rolled into a cylinder of height $10$ cm so that the $6\pi$ cm edges meet without overlap. What is the volume of this cylinder?

**a.** $30\pi\,\text{cm}^3$

**b.** $60\pi\,\text{cm}^3$

**c.** $90\pi\,\text{cm}^3$

**d.** $360\pi\,\text{cm}^3$

**34** [Intermediate] • [MCQ] • Target: Surface-Area-to-Volume Ratio of Similar Solids

The ratio of the surface areas of two cubes is $16:49$. What is the ratio of their volumes?

**a.** $4:7$

**b.** $16:49$

**c.** $64:343$

**d.** $256:2401$

**35** [Intermediate] • [MCQ] • Target: Parallelogram Area via Trigonometry

A parallelogram has adjacent sides of lengths $10$ and $14$, and the acute angle between them is $30^\circ$. What is the area of the parallelogram?

**a.** $35$

**b.** $70$

**c.** $70\sqrt{3}$

**d.** $140$

**36** [Intermediate] • [MCQ] • Target: Volume of a Conical Frustum

A frustum is formed by slicing off the top third (by height) of a right circular cone of volume $135\,\text{cm}^3$ with a plane parallel to its base. What is the volume of the remaining bottom frustum?

**a.** $90\,\text{cm}^3$

**b.** $115\,\text{cm}^3$

**c.** $120\,\text{cm}^3$

**d.** $130\,\text{cm}^3$

**37** [Intermediate] • [MCQ] • Target: Total Surface Area of a Solid Hemisphere

A solid hemisphere of radius $R$ has total surface area (including its flat circular base) equal to $75\pi$. What is $R$?

**a.** $3$

**b.** $5$

**c.** $5\sqrt{3}$

**d.** $15$

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## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Isoperimetric Optimization via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Optimization)

Among all rectangles with a fixed perimeter of $48\,\text{cm}$, what is the maximum possible area?

**a.** $120\,\text{cm}^2$

**b.** $135\,\text{cm}^2$

**c.** $144\,\text{cm}^2$

**d.** $256\,\text{cm}^2$

**39** [Challenging] • [MCQ] • Target: Archimedes' Sphere-to-Cylinder Ratio • Cross-Lesson: Topic 5.01 (Ratios)

A sphere is inscribed in a right circular cylinder so that the sphere touches the top base, bottom base, and lateral wall. What is the ratio of the volume of the sphere to the volume of the cylinder?

**a.** $1:2$

**b.** $2:3$

**c.** $3:4$

**d.** $4:3$

**40** [Challenging] • [MCQ] • Target: Volume from Pairwise Face Areas • Cross-Lesson: Topic 4.01 (Radicals & Algebraic Products)

In a rectangular prism, the areas of three mutually adjacent faces are $12\,\text{cm}^2$, $15\,\text{cm}^2$, and $20\,\text{cm}^2$. What is the volume of the prism?

**a.** $47\,\text{cm}^3$

**b.** $60\,\text{cm}^3$

**c.** $120\,\text{cm}^3$

**d.** $180\,\text{cm}^3$

**41** [Challenging] • [MCQ] • Target: Coordinate Solids of Revolution • Cross-Lesson: Topic 1.01 (Coordinate Geometry)

A region in the first quadrant bounded by $x = 0$, $y = 0$, $x = 4$, and $y = 3$ is revolved $360^\circ$ about the $y$-axis. What is the volume of the resulting solid?

**a.** $36\pi$

**b.** $48\pi$

**c.** $64\pi$

**d.** $144\pi$

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**42** [Challenging] • [MCQ] • Target: Algebraic Surface-to-Volume Equating • Cross-Lesson: Topic 4.01 (Polynomial Equations)

The surface area of a cube is numerically equal to twice its volume. What is the length of the space diagonal of the cube?

**a.** $3$

**b.** $3\sqrt{2}$

**c.** $3\sqrt{3}$

**d.** $6\sqrt{3}$

**43** [Challenging] • [MCQ] • Target: Tilted Cylinder Volume Symmetry • Cross-Lesson: Topic 7.01 (45-45-90 Triangles)

A right circular cylinder of radius $6\,\text{cm}$ and height $15\,\text{cm}$ is initially completely full of water, and is then tilted at a $45^\circ$ angle until water stops spilling out. What volume of water was spilled?

**a.** $108\pi\,\text{cm}^3$

**b.** $180\pi\,\text{cm}^3$

**c.** $216\pi\,\text{cm}^3$

**d.** $324\pi\,\text{cm}^3$

**44** [Challenging] • [MCQ] • Target: Composite Area of Polygons & Circular Sectors • Cross-Lesson: Topic 9.01 (Circular Sectors)

An equilateral triangle of side length $6$ has circular sectors of radius $3$ centered at each of its three vertices removed. What is the area of the remaining central region?

**a.** $9\sqrt{3} - \frac{9\pi}{2}$

**b.** $9\sqrt{3} - 9\pi$

**c.** $18\sqrt{3} - \frac{9\pi}{2}$

**d.** $36\sqrt{3} - 9\pi$

**45** [Challenging] • [MCQ] • Target: Percentage Scaling of Surface Area • Cross-Lesson: Topic 5.01 (Percentages)

If the edge length of a cube is increased by $10\%$, by what percentage does its total surface area increase?

**a.** $10\%$

**b.** $20\%$

**c.** $21\%$

**d.** $33.1\%$

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